



WHEN

March 20, 2026
Doors open at 7:00pm
Meeting starts at 7:30pm
Lecture starts at 8:00pm

WHERE

Unitarian Church
1893 North Vasco Road
Livermore, CA 945571
and via Zoom

TVS QR CODE



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THIS MONTH'S SPEAKER

Herbig-Haro Objects: The Smoking Guns of Early Star Formation

Lisa Prato

As protostars coalesce in the cold dark cores of molecular clouds, they spin up to rapid rotation rates while the remnant material from star formation collapses into dusty gaseous circumstellar disks. The interactions between disks and strong young stars' magnetic fields spawn energetic gas jets and outflows that subsequently slam into the interstellar medium, exciting the cloud material. Thus, observations of these shocked regions, the Herbig-Haro objects, are superb tracers of early star formation. I will describe recent advances by my team in identifying, cataloging, and understanding these objects in the nearby star forming regions.



HH 46/47. NASA, ESA, CSA; Image Processing: Joseph DePasquale (STScI), Anton Koekemoer (STScI)

Lisa Prato grew up in the Boston area and attended the University of Massachusetts, Amherst, where she completed undergraduate degrees in Astronomy and in English and discovered the world of scientific research. After another year at UMass as a non-degree graduate student, Prato moved to Colombia, South America, and worked for 3 years as a professor in the Department of Physics at the Universidad Industrial de Santander in the city of Bucaramanga, teaching classes in astronomy and astrophysics and working toward a master's degree in Physics. Prato returned to the US to complete a PhD in Astronomy at the State University of New York at Stony Brook, followed by two postdoctoral positions at UCLA, one with the Nobel Laureate Dr. Andrea Ghez and one with Dr. Ian McLean. Prato was a Tenured Astronomer on the science faculty at Lowell Observatory for 21 years and is currently a Research Professor at Northern Arizona University; she works with postdocs, undergraduates, interns, and graduate students on young binary stars and planet formation.

NEWS AND NOTES

Meeting Dates

Club Meeting	Board Meeting	PrimeFocus Deadline
March 20	March 23	March 7
April 17	April 20	April 4
May 15	May 18	May 9

Money Matters

As of the last Treasurer's Report on 12/22/25, our club's account balance is \$49,833.30, this includes \$12,569.51 in the H2O Rebuild fund. No treasury update during 2/23/26 board meeting..

TVS Welcomes New Members

TVS welcomes new members Amanda Brown and Aditya Jagtiani. Please say hello and chat with them during our meetings.

2026 TVS Club Star Party Schedule

Save the dates for the 2025 Club Star Parties. Del Valle star parties are also public outreach events. They are jointly hosted with the EBRPD and held at the Arroyo Staging Area (Coords: 37.6196638, -121.7528899). The public is invited for the first 1.5-2 hours, while club members can stay the remainder of the night.

No events currently scheduled for Del Valle.

Tesla Vintners star parties are open to only club members and their guests. These star parties end at midnight, but participants can leave earlier, should they wish.

April 11: Tesla Vintner's Star Party, 5143 Tesla Rd., Livermore. Set-up at 7:30pm, Observing 8:00-11:30pm.

June 6: Tesla Vintner's Star Party, 5143 Tesla Rd., Livermore. Set-up at 8:00pm, Observing 8:30-11:30pm.

August 8: Tesla Vintner's Star Party, 5143 Tesla Rd., Livermore. Set-up at 7:30pm, Observing 8:00-Midnight.

October 10: Tesla Vintner's Star Party, 5143 Tesla Rd., Livermore. Set-up at 6:30pm, Observing 7:00pm-11-30pm.

H2O Open House star parties are open to only club members and their guests. The open house ends at midnight, and all participants are encouraged to stay the duration. The drive to H2O takes about 1 hour, and the caravan leaves promptly from the corner of Mines and Tesla Rds. No gas stations are available on the route, so be prepared. Admission is \$3/car-bring exact change. H2O is a primitive site with two porta-potties. Bring water, food, and warm clothing, as needed. Red flashlights are to be used so observers can preserve their night vision.

May 9: H2O Open House, 5:00pm caravan to H2O PROMPTLY leaves the corner of Mines and Tesla Rds., Livermore. Observing until 11:30pm. Meeting times are tentative.

September 5: H2O Open House, 5:00pm caravan to H2O PROMPTLY leaves the corner of Mines and Tesla Rds., Livermore. Observing until 11:30pm. Meeting times are tentative.

March 21: Public Star Party at Sunol Regional Wilderness, 1895 Geary Road, Sunol. Set-up 7:00pm, Observing from 7:45-8:45pm.

April 4: Cub Scout star party at Del Valle Arroyo Road Staging Area, Del Valle Arroyo Road Staging Area, Arroyo Road. Set-up 6:30pm, Observing from 7:30-9:30pm

May 23: Public Star Party at Sunol Regional Wilderness, 1895 Geary Road, Sunol. Set-up 7:30pm, Observing from 8:00-9:30pm.

CALENDAR OF EVENTS



March 20, 21, 27,28, April 3, 4, 10,11, 17, 18, 7:30-10:30pm	
What	Free Telescope Viewing
Who	Chabot Staff
Where	Chabot Space and Science Center, 10000 Skyline Blvd. Oakland, CA 94619
Cost	Free

Join Chabot astronomers on the Observatory Deck for a free telescope viewing! Weather permitting, this is a chance to explore stars, planets and more through Chabot's historic telescopes. Chabot's three large historic telescopes offer a unique way to experience the awe and wonder of the Universe. Three observatory domes house the Center's 8-inch (Leah, 1883) and 20-inch (Rachel, 1916) refracting telescopes, along with a 36-inch reflecting telescope (Nellie, 2003).

Are the skies clear for viewing tonight? Viewing can be impacted by rain, clouds, humidity and other weather conditions. Conditions can be unique to Chabot because of its unique location in Joaquin Miller Park. Before your visit, check out the Weather Station to see the current conditions at Chabot.

For more information, see <https://chabotspace.org/events/events-listing/> <https://eastbayastro.org/chabot-telescope-status/>



Leo Triplet by Erny Epley

WHATS UP

Adapted from Sky and Telescope

All times are Pacific Standard Time

March 2026

20	Fri	Spring begins at 7:46 AM; See Venus just above the western horizon with the thin sliver of the Moon
25	Wed	Moon is at first quarter; Looking west see the moon $6\frac{1}{2}^{\circ}$ lower right of Jupiter.
29	Sun	Moon is trails Regulus by just 3° as they rise at dusk in the south east

April 2026

2	Thu	Full Moon
6	Mon	At dawn, looking southeast, Antares is just $4\frac{1}{2}^{\circ}$ upper left of the Moon
10	Fri	Moon is at last quarter
17	Fri	New Moon

OFFICERS AND VOLUNTEER POSITIONS

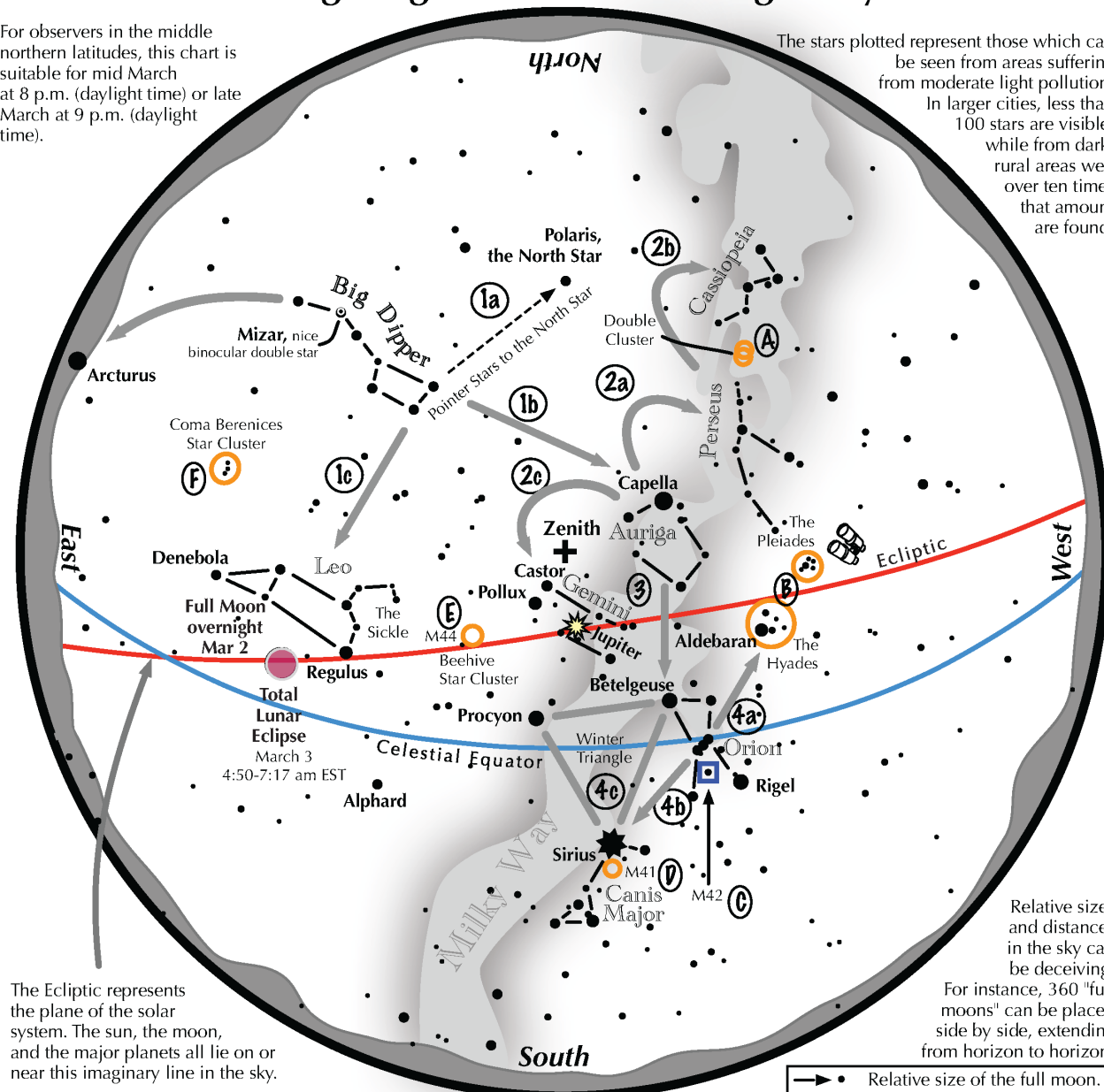
Officers	Volunteer Positions	Newsletter	Potluck Coordinator
President Eric Dueltgen president@trivalleystargazers.org	Astronomical League Rep. Don Dossa alrep@trivalleystargazers.org	Scott Schneider (Editor) newsletter@trivalleystargazers.org	OPEN potluck@trivalleystargazers.org
Vice-President Aris Pope vice_president@trivalleystargazers.org	Club Star Party Coordinator Johnathan Bailey coordinator@trivalleystargazers.org	Webmaster Swaroop Shere webmaster@trivalleystargazers.org	Program Coordinator Ron Kane programs@trivalleystargazers.org
Treasurer Ernest Epley treasurer@trivalleystargazers.org	Del Valle Coordinator David Wright delvalle@trivalleystargazers.org	Night Sky Network Rep. Ross Gaunt nnsn@trivalleystargazers.org	Publicity and Fundraising OPEN publicity@trivalleystargazers.org
Secretary Dave Lackey secretary@trivalleystargazers.org	Historian OPEN historian@trivalleystargazers.org	H2O Observatory Director / Rebuild Chairman Chuck Grant H2O@trivalleystargazers.org	Refreshment Coordinator OPEN
Past President Ron Kane past_president@trivalleystargazers.org	Librarian Ron Kane librarian@trivalleystargazers.org	Observing Program Coordinator Ron Kane awards@trivalleystargazers.org	Web and Email www.trivalleystargazers.org info@trivalleystargazers.org
	Loaner Scope Manager Ron Kane telescopes@trivalleystargazers.org	Outreach Coordinator Johnathan Bailey outreach@trivalleystargazers.org	TVS E-Group To Join the TVS E-Group just send an email to TVS at info@trivalleystargazers.org asking to join the group. Make sure you specify the email address you want to use to read and post to the group.

NAVIGATING THE NIGHT SKY FOR MARCH

Navigating the mid March Night Sky

For observers in the middle northern latitudes, this chart is suitable for mid March at 8 p.m. (daylight time) or late March at 9 p.m. (daylight time).

The stars plotted represent those which can be seen from areas suffering from moderate light pollution. In larger cities, less than 100 stars are visible, while from dark, rural areas well over ten times that amount are found.



The Ecliptic represents the plane of the solar system. The sun, the moon, and the major planets all lie on or near this imaginary line in the sky.

Relative sizes and distances in the sky can be deceiving. For instance, 360 "full moons" can be placed side by side, extending from horizon to horizon.

→ • Relative size of the full moon.

Navigating the March night sky: Simply start with what you know or with what you can easily find.

- 1 Above the northeast horizon rises the Big Dipper. Draw a line from its two end bowl stars upwards to the North Star. Its top bowl stars point west to Capella in Auriga, nearly overhead. Leo reclines below the Dipper's bowl.
- 2 From Capella jump northwestward along the Milky Way to Perseus, then to the "W" of Cassiopeia. Next jump southeastward from Capella to the twin stars of Castor and Pollux in Gemini.
- 3 Directly south of Capella stands the constellation of Orion with its three Belt Stars, its bright red star Betelgeuse, and its bright blue-white star Rigel.
- 4 Use Orion's three Belt stars to point northwest to the red star Aldebaran and the Hyades star cluster, then to the Pleiades star cluster. Travel southeast from the Belt stars to the brightest star in the night sky, Sirius. It is a member of the Winter Triangle.

Binocular Highlights

A: Between the "W" of Cassiopeia and Perseus lies the Double Cluster. **B:** Examine the stars of the Pleiades and Hyades, two naked eye star clusters. **C:** M42 in Orion is a star forming nebula. **D:** Look south of Sirius for the star cluster M41. **E:** M44, a star cluster barely visible to the naked eye, lies to the southeast of Pollux. **F:** Look high in the east for the loose star cluster of Coma Berenices.



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Tri-Valley Stargazers
P.O. Box 2476
Livermore, CA 94551
www.trivalleystargazers.org

Tri-Valley Stargazers Membership Application

Contact information:

Name: _____ Phone: _____

Street Address: _____

City, State, Zip: _____

Email Address: _____

Status (select one): _____ New member _____ Renewing or returning member

Membership category (select one): Membership term is for one calendar year, January through December.

_____ Student member (\$10). Must be a full-time high-school or college student.

_____ Regular member (\$30).

Hidden Hill Observatory Access (optional): Must be 18 or older.

_____ One-time key deposit (\$20). This is a refundable deposit for a key to H2O. New key holders must first hear an orientation lecture and sign a usage agreement form before using the observing site.

_____ Annual access fee (\$10). You must also be a key holder to access the site.

Donation (optional):

_____ Tax-deductible contribution to Tri-Valley Stargazers

Total enclosed: \$ _____

Member agrees to hold Tri-Valley Stargazers, and any cooperating organizations or landowners, harmless from all claims of liability for any injury or loss sustained at a TVS function. TVS will not share information with anyone except as detailed in our Privacy Policy (<http://www.trivalleystargazers.org/privacy.shtml>).

Mail this completed form along with a check to: Tri-Valley Stargazers, P.O. Box 2476, Livermore, CA 94551.